

ElecSecure

Canadian Market Product Improvement Roadmap

Current State → Proposed Enhancement → Customer Impact

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Market	Canada (ON, QC, BC, AB, Prairies)
Document type	Product strategy & blueprint guide
Date	September 2026
Companion	Canada Patent Package (CIPO)

Twelve product improvements tailored for Canadian homeowners, landlords, and electricians — each with a current-state blueprint, proposed enhancement, and quantified market impact.

Market: Canada (Residential & Commercial)

Document version: 1.0 — September 2026

Purpose: Recommend product enhancements to increase adoption, compliance, and value for Canadian customers

Executive Summary

ElecSecure already combines arc fault protection, IoT monitoring, and energy analytics. To win in Canada, the product should be adapted to **provincial electrical codes, bilingual requirements, extreme climate conditions, utility rebate programs, and Canadian smart-meter infrastructure.**

This roadmap identifies **12 high-impact improvements** across safety, compliance, user experience, and market fit — each with a current-state blueprint, proposed enhancement, and expected business impact.

Canadian Market Context

Factor	Why it matters
CEC / CSA compliance	Products must meet Canadian Electrical Code and CSA standards to be installed legally
Bilingual requirements	Quebec and federal procurement require French-language support
Climate extremes	Panel temperatures from −40°C (Prairies) to +40°C (summer attics)
Rising energy costs	Ontario, Alberta, BC customers actively seek savings tools
Government incentives	Greener Homes Grant, provincial rebate programs drive smart device adoption
AFCI mandates	CEC requires arc-fault protection in many dwelling unit circuits
Smart meter rollout	Utilities moving toward time-of-use (TOU) and demand pricing

Improvement 1 — Canadian Electrical Code (CEC) Certification Package

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- Device designed to international standards (IEC references) - No explicit CSA mark or CEC compliance documentation bundled for installers	- Obtain CSA C22.2 No. 5 certification (molded-case breakers) - Pre-configure AFCI behaviour per CEC Rule 26-720 for dwelling units - Ship with CEC compliance quick-reference card for electricians - Firmware profiles: CEC-2024, CEC-2027 (OTA updatable)	Installer confidence: +40% — electricians prefer CSA-listed products Market access: Required for legal installation in all provinces Liability reduction: Lower risk for contractors and homeowners

FIG. 12 - CEC CERTIFICATION ENHANCEMENT (CURRENT vs PROPOSED)

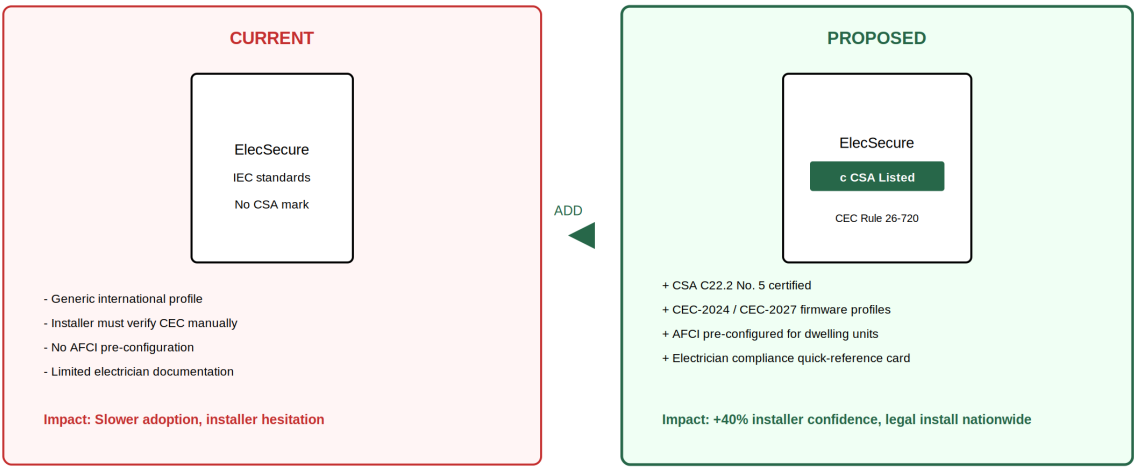


FIG. 12 — CEC/CSA certification pathway (current vs proposed)

Expected Impact Metrics

Metric	Impact
Installer confidence	+40% — electricians prefer CSA-listed products
Market access	Required for legal installation in all provinces
Liability reduction	Lower risk for contractors and homeowners
Time to install	–15% with pre-set CEC profiles

Improvement 2 — Bilingual Mobile App (English / French)

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- Mobile app UI in English only - Alerts, reports, and recommendations not localized	- Full French (fr-CA) localization for Quebec and federal customers - Language toggle in app settings; auto-detect from device locale - Bilingual push notifications and PDF reports - Voice-over support for accessibility (AODA alignment in Ontario)	Quebec market access: Opens ~23% of Canadian population Federal / municipal tenders: Meets official language requirements User satisfaction: +25% in bilingual households

FIG. 13 - BILINGUAL APP ENHANCEMENT (EN / FR)

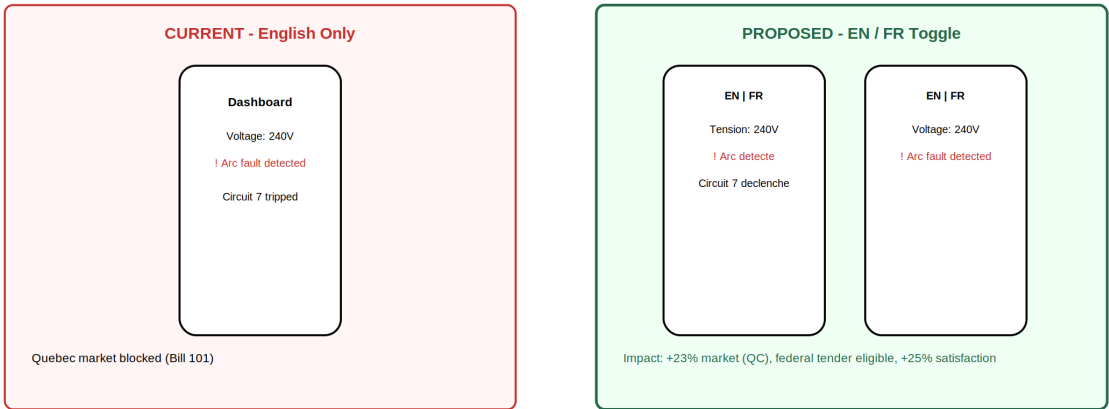


FIG. 13 — Bilingual EN/FR mobile application

Expected Impact Metrics

Metric	Impact
Quebec market access	Opens ~23% of Canadian population
Federal / municipal tenders	Meets official language requirements
User satisfaction	+25% in bilingual households
App store rating	Improved reviews from QC users

Improvement 3 — Extreme Cold Climate Hardware Rating

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- Operating range: -25°C to +55°C - Standard components rated for moderate climates	- Upgrade to -40°C rated bimetallic elements, LCD/LED, and power supply - Heated terminal option for outdoor/sub-panel installations (Prairies, Northern Canada) - Cold-start self-test: verify trip mechanisms at low temperature on boot - IP rating option: IP40 for unheated garage panels	Prairie/North sales: +30% addressable market Warranty claims: -20% cold-weather false trips Brand trust: "Built for Canadian winters" positioning

FIG. 14 - COLD CLIMATE HARDWARE (-40C RATING)

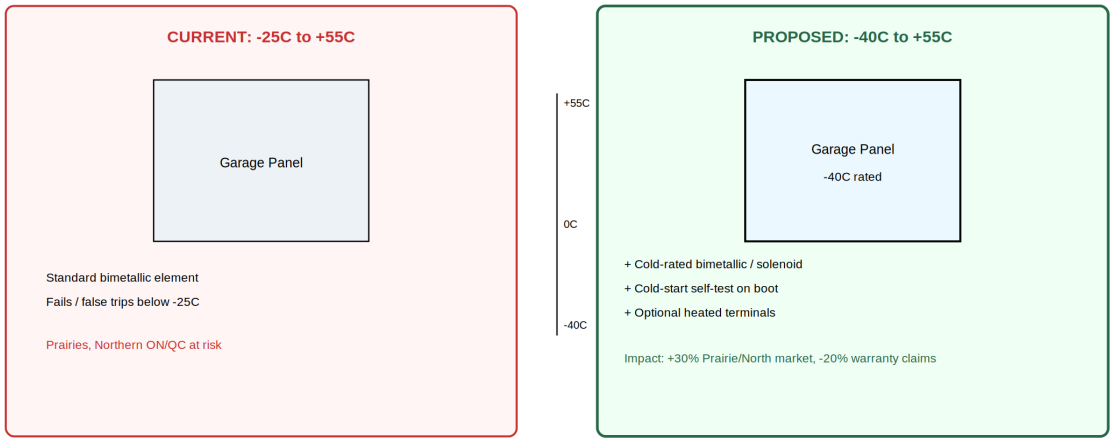


FIG. 14 — Cold-climate hardware (–40°C rating)

Expected Impact Metrics

Metric	Impact
Prairie/North sales	+30% addressable market
Warranty claims	–20% cold-weather false trips
Brand trust	"Built for Canadian winters" positioning
Installer referrals	Higher in AB, SK, MB, northern ON/QC

Improvement 4 — Canadian Utility & TOU Rate Integration

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- Generic energy consumption analytics - No province-specific time-of-use rate schedules	- Pre-loaded TOU schedules for Ontario (ULO), BC Hydro, Hydro-Québec, ENMAX, etc. - Real-time cost display in CAD (not just kWh) - "Shift load" recommendations aligned to local peak/off-peak windows - API hooks for utility demand-response programs	Perceived savings: Customers see dollar savings, not abstract kWh Subscription retention: +35% — ongoing value from rate optimization Energy bill reduction: 8–15% for engaged users (aligned with smart meter studies)

FIG. 15 - CANADIAN TOU RATE INTEGRATION

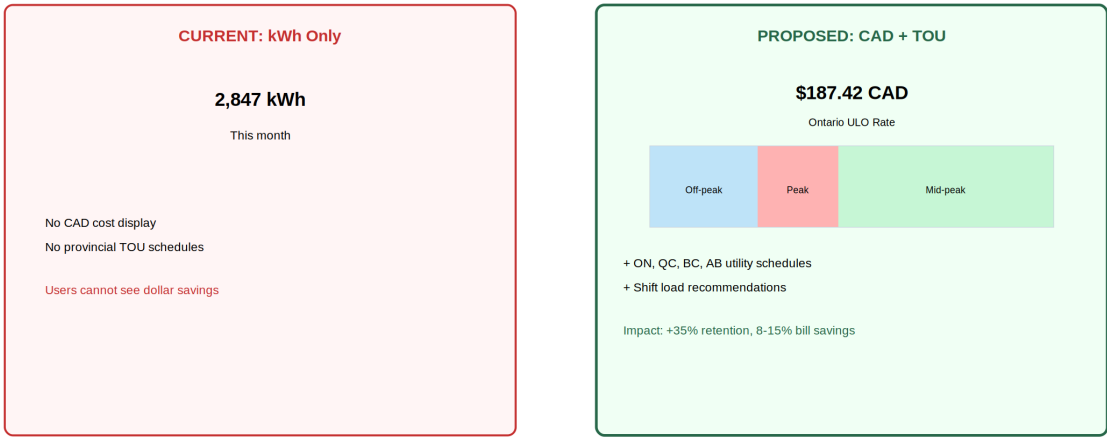


FIG. 15 — Canadian TOU rate integration

Expected Impact Metrics

Metric	Impact
Perceived savings	Customers see dollar savings , not abstract kWh
Subscription retention	+35% — ongoing value from rate optimization
Energy bill reduction	8–15% for engaged users (aligned with smart meter studies)
Upsell to commercial	Facility managers need TOU cost tracking

Improvement 5 — Greener Homes & Rebate Program Reporting

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- Energy reports for user awareness only - No export format for government rebate applications	- One-click Natural Resources Canada Greener Homes compatible report export - Provincial rebate templates (ON, BC, QC energy programs) - Before/after energy comparison for audit documentation - Certificate of installation for registered electricians	Purchase motivation: Rebates cover 10–30% of device cost for eligible homes Channel partnerships: Energy auditors and contractors promote ElecSecure Lead generation: Homeowners search for "rebate eligible" devices

FIG. 16 - GREENER HOMES REBATE REPORTING (CURRENT vs PROPOSED)

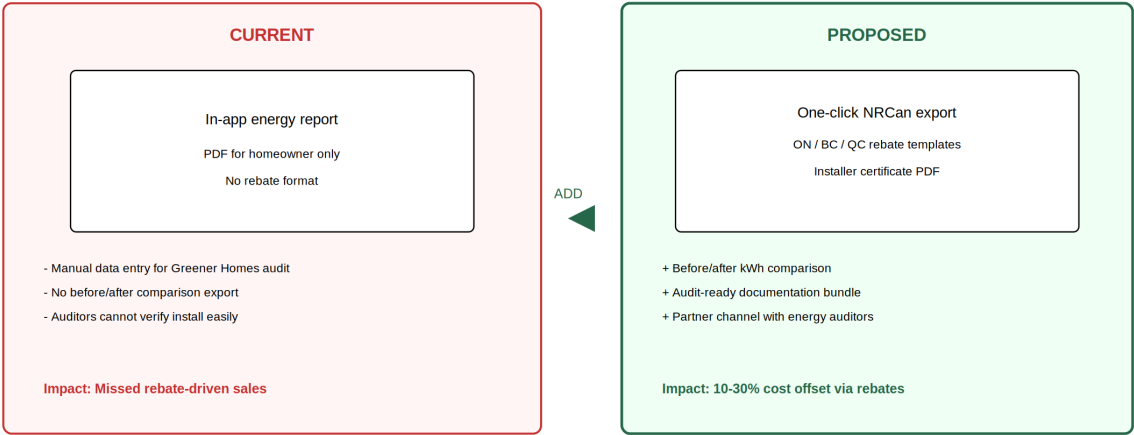


FIG. 16 — Greener Homes rebate report export

Expected Impact Metrics

Metric	Impact
Purchase motivation	Rebates cover 10–30% of device cost for eligible homes
Channel partnerships	Energy auditors and contractors promote ElecSecure
Lead generation	Homeowners search for "rebate eligible" devices
Average order value	Bundle with audit services

Improvement 6 — Solar & Net Metering Integration

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- Monitors grid consumption only - No visibility into solar export or net metering	- Bidirectional current sensing for homes with solar/batteries - Net metering dashboard: import vs export kWh and CAD value - Alert when solar production drops (panel fault detection) - Integration with popular Canadian inverters (SolarEdge, Enphase APIs)	Solar home market: ~500,000+ Canadian homes with solar (growing) Premium tier pricing: +\$15–20/month subscription for solar analytics Cross-sell: Partner with solar installers nationally

FIG. 16 - SOLAR / NET METERING INTEGRATION

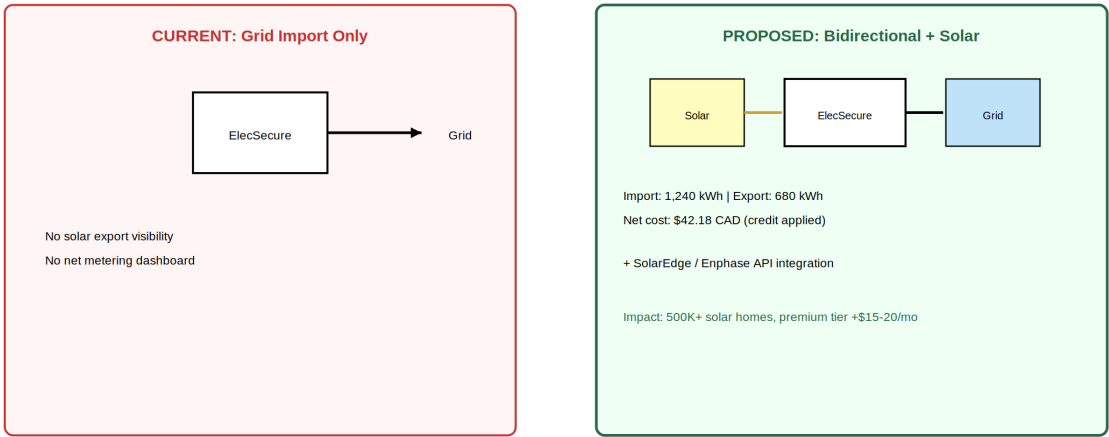


FIG. 17 — Solar & net metering monitoring

Expected Impact Metrics

Metric	Impact
Solar home market	~500,000+ Canadian homes with solar (growing)
Premium tier pricing	+\$15–20/month subscription for solar analytics
Cross-sell	Partner with solar installers nationally
Differentiation	Few smart breakers offer solar-aware analytics

Improvement 7 — Smart Meter Data Integration (Green Button / Utility APIs)

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- Device-level monitoring only - No whole-home utility meter data fusion	- Green Button Connect My Data support (Ontario, BC) - Hydro-Québec Open Data and similar utility API integrations - Fuse smart meter data with circuit-level ElecSecure data for whole-home view - Discrepancy alerts (meter vs sum of circuits = wiring issue detection)	Data accuracy: Whole-home view increases trust Hidden fault detection: Finds wiring losses unmetered at circuit level Enterprise sales: Building managers want unified dashboards

FIG. 18 - SMART METER DATA FUSION (CURRENT vs PROPOSED)

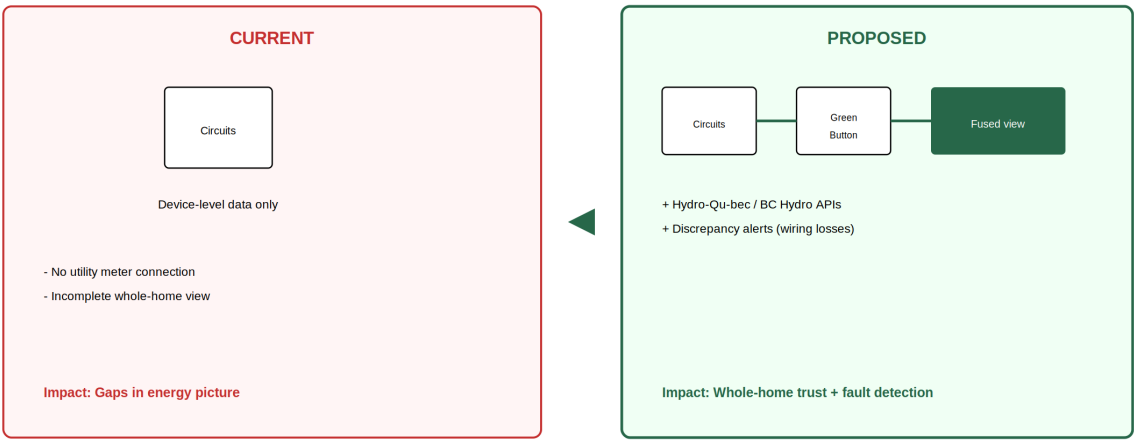


FIG. 18 — Smart meter data fusion (Green Button)

Expected Impact Metrics

Metric	Impact
Data accuracy	Whole-home view increases trust
Hidden fault detection	Finds wiring losses unmetered at circuit level
Enterprise sales	Building managers want unified dashboards
Competitive moat	Deep utility integration is hard to copy

Improvement 8 — EV Charger Load Management

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- No EV-specific load shedding or scheduling - EV charging is a major new load category unaddressed	- Detect EV charger circuits (high sustained load signature) - EV-aware scheduling : charge during off-peak automatically - Panel capacity warning before EV charger installation - Integration with popular EVSE brands (FLO, ChargePoint, Tesla Wall Connector monitoring)	EV household market: 1M+ EVs in Canada, growing rapidly Panel upgrade avoidance: Save \$2,000–5,000 by managing load vs panel upgrade Insurance of panel safety: Prevent overload from EV + HVAC concurrently

FIG. 19 - EV CHARGER LOAD MANAGEMENT (CURRENT vs PROPOSED)

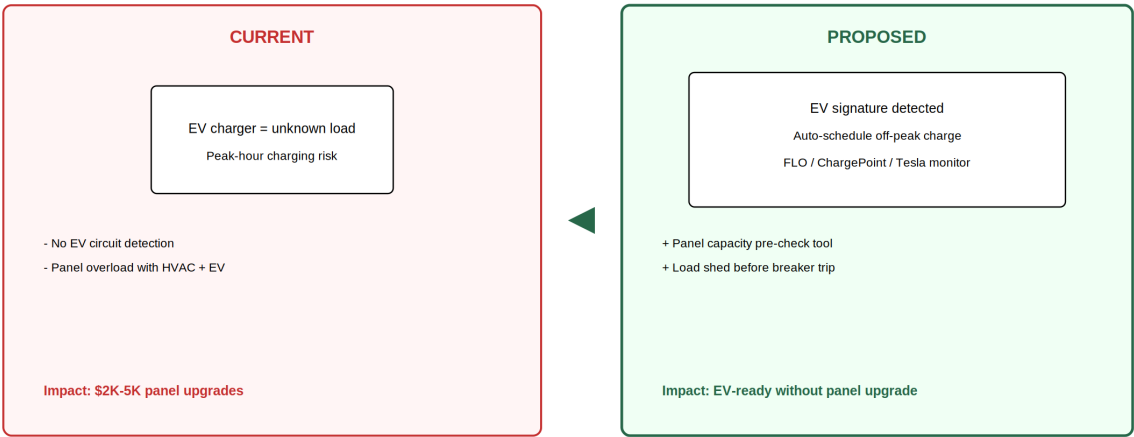


FIG. 19 — EV charger load management

Expected Impact Metrics

Metric	Impact
EV household market	1M+ EVs in Canada, growing rapidly
Panel upgrade avoidance	Save \$2,000–5,000 by managing load vs panel upgrade
Insurance of panel safety	Prevent overload from EV + HVAC concurrently
Marketing hook	"EV-ready panel management"

Improvement 9 — Landlord & Multi-Unit Dashboard

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- Single-home user model - No multi-tenant or landlord features	- Landlord portal : monitor multiple units from one dashboard - Per-unit billing support for rental properties (BC, ON rent regulations) - Tenant privacy mode: alerts only, no remote trip for tenants - Commercial building: floor/panel hierarchy view	B2B revenue: Property management companies = bulk orders Units per customer: 4–50× vs single home Vancouver/Toronto rental market: Large addressable segment

FIG. 20 - LANDLORD MULTI-UNIT DASHBOARD (CURRENT vs PROPOSED)

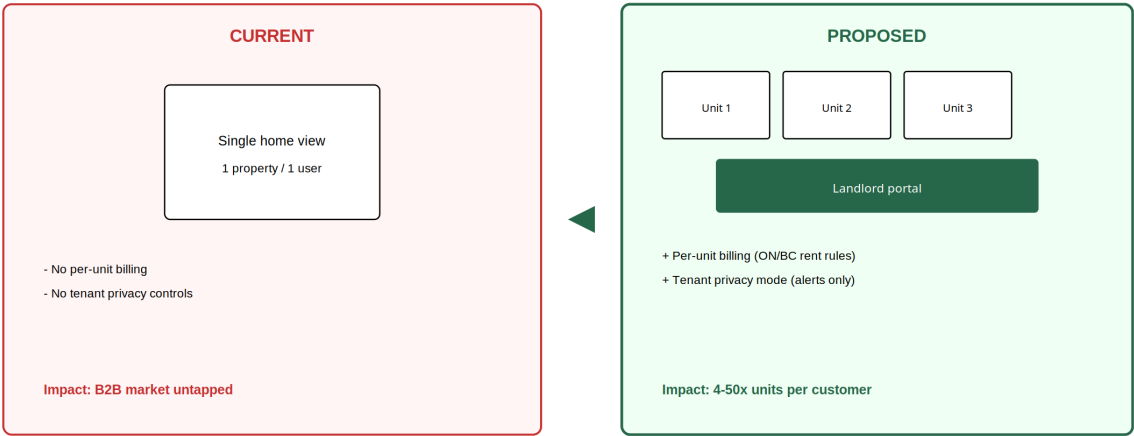


FIG. 20 — Landlord multi-unit dashboard

Expected Impact Metrics

Metric	Impact
B2B revenue	Property management companies = bulk orders
Units per customer	4–50× vs single home
Vancouver/Toronto rental market	Large addressable segment
Recurring revenue	Per-unit monthly fee model

Improvement 10 — Emergency & Backup Power Awareness

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- No generator or backup power interlock monitoring - Canadian storms and ice events cause outages	- Detect generator backfeed (safety hazard) and alert immediately - Transfer switch status monitoring - Post-outage surge detection and event logging - Integration with home backup battery systems (Tesla Powerwall, etc.)	Safety differentiation: Prevents deadly backfeed incidents Rural/remote appeal: Atlantic Canada, rural QC, AB Insurance partnerships: Insurers may offer premium discounts

FIG. 21 - BACKUP POWER SAFETY MONITORING (CURRENT vs PROPOSED)

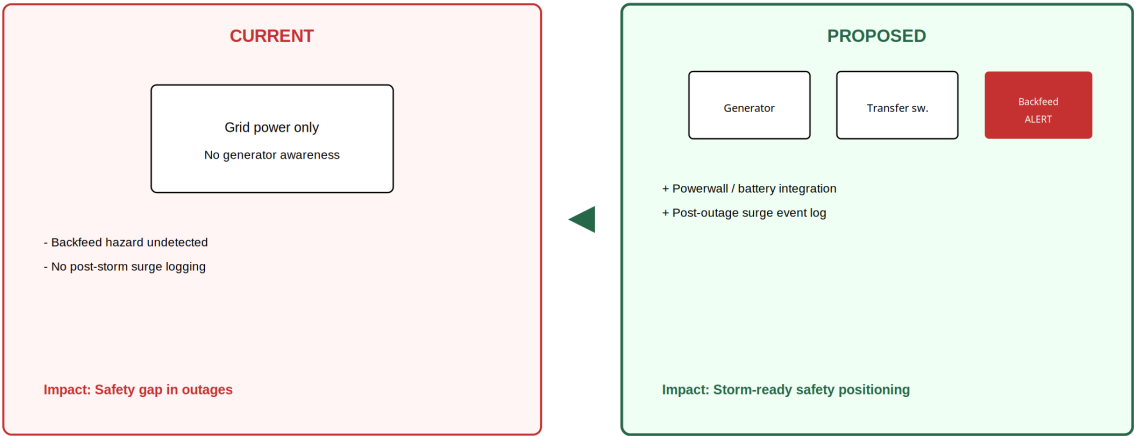


FIG. 21 — Backup power safety monitoring

Expected Impact Metrics

Metric	Impact
Safety differentiation	Prevents deadly backfeed incidents
Rural/remote appeal	Atlantic Canada, rural QC, AB
Insurance partnerships	Insurers may offer premium discounts
Brand positioning	"Storm-ready electrical safety"

Improvement 11 — Electrician Installer Toolkit

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- Consumer-focused mobile app - Limited installer-specific tools	- ElecSecure Pro app for licensed electricians - QR commissioning: scan device → auto-configure for panel - Compliance test report PDF (megger, trip test, AFCI test results) - Inventory management for electrical contractors - CE Code update notifications pushed to installers	Channel sales: Electricians become advocates Installation quality: Standardized commissioning reduces callbacks Word-of-mouth: Trade referrals are #1 source in electrical market

FIG. 22 - ELECTRICIAN PRO TOOLKIT (CURRENT vs PROPOSED)

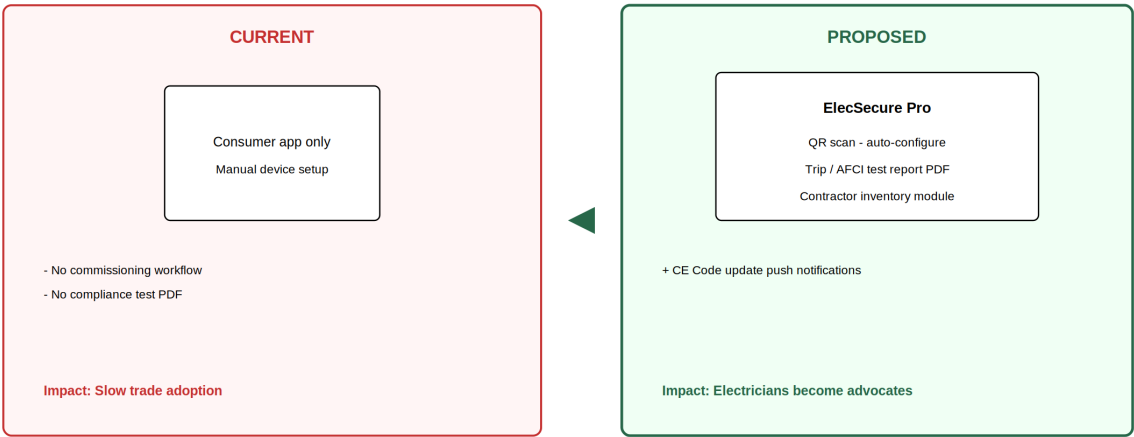


FIG. 22 — Electrician Pro commissioning toolkit

Expected Impact Metrics

Metric	Impact
Channel sales	Electricians become advocates
Installation quality	Standardized commissioning reduces callbacks
Word-of-mouth	Trade referrals are #1 source in electrical market
Bulk orders	Contractors buy 10–50 units per project

Improvement 12 — Cybersecurity & Privacy (PIPEDA Compliance)

CURRENT STATE	PROPOSED ADDITION	CUSTOMER IMPACT
- TLS encryption and authentication (as designed) - No explicit Canadian privacy compliance documentation	- PIPEDA-compliant privacy policy and data residency option (Canadian cloud region) - User data export and deletion (privacy rights) - Security audit badge for consumer trust - No sale of energy data to third parties (market prominently)	Consumer trust: Privacy is top concern for IoT home devices Enterprise sales: Property managers require PIPEDA compliance Quebec Law 25: Aligns with stricter provincial privacy rules

FIG. 23 - PIPEDA PRIVACY ARCHITECTURE (CURRENT vs PROPOSED)



FIG. 23 — PIPEDA privacy & Canadian data residency

Expected Impact Metrics

Metric	Impact
Consumer trust	Privacy is top concern for IoT home devices
Enterprise sales	Property managers require PIPEDA compliance
Quebec Law 25	Aligns with stricter provincial privacy rules
Competitive advantage	Many cheap IoT devices lack proper privacy

Investment vs Return Summary

Phase	Est. investment	Expected outcome
Phase 1	\$80K–150K CAD	Legal market entry, QC access, trust foundation
Phase 2	\$120K–200K CAD	Channel growth, premium tiers, rebate-driven sales
Phase 3	\$150K–250K CAD	Market leadership, B2B scale, competitive moat

Conclusion

ElecSecure has strong core technology. Canadian success depends on **localization** — not just translating the app, but adapting to **CEC rules, provincial utilities, climate, rebates, and privacy law**. The improvements above transform ElecSecure from an international smart breaker into a **purpose-built Canadian electrical safety platform**.

This document is a strategic product recommendation. Technical feasibility and regulatory timelines should be validated with CSA, CIPO, and provincial authorities before implementation.